

Yuanzhuo Sun

Chapel Hill, NC | (919) 593-1703 | syuanzhu@unc.edu

Research Interests

Applied mathematics; experimental and computational fluid dynamics; density-stratified flows; inverse problems and optical diagnostics; scientific computing and numerical methods; stochastic processes; population genealogies; and mathematical biology.

Education

University of North Carolina at Chapel Hill

B.S. in Statistics and Analytics; B.S. in Data Science; Minor in Mathematics

Expected May 2027

Chapel Hill, NC

- GPA: 3.50/4.00.
- Selected coursework: Advanced Calculus, Differential Equations, Numerical Methods, Probability, Optimization, Stochastic Modeling, Machine Learning, and Statistical Methods.

Research Experience

Undergraduate Research Assistant — Joint Fluids Lab

Feb. 2025 – Present

Advisors: Richard M. McLaughlin and Roberto Camassa, UNC Department of Mathematics

Chapel Hill, NC

- Conduct experimental and computational research on density-stratified miscible fluids, hydrodynamic instabilities, and optical diagnostics under NSF- and ONR-supported research projects.
- Serve as a primary undergraduate experimentalist, contributing to experimental design, instrumentation, data acquisition, image analysis, and comparison with mathematical models.

Optical Inference of Density Stratification in Miscible Fluids via Refractive Signature Analysis

2026 SURF Project

- Designed and built an automated two-axis optical scanning system controlling laser height and incident angle, synchronized with high-resolution image acquisition.
- Collected and calibrated thousands of refracted laser trajectories through stably stratified salt-water experiments across controlled initial conditions and density gradients.
- Developed image-processing and geometric-calibration workflows to extract physical ray trajectories, including strongly curved and preliminary folding/overturning paths.
- Compare measured trajectories with nonlinear ODE-based ray models toward a non-invasive inverse method for reconstructing hidden density profiles from optical measurements.

Interfacial Flow Patterns and Obstacle-Induced Instabilities in a Density-Stratified Hele–Shaw Cell

ONR-Supported Project, Completed 2026

- Investigated how density stratification, obstacle geometry, and gap confinement modify shear layers, wake formation, and flow symmetry in quasi-two-dimensional Hele–Shaw flows.
- Conducted controlled stratified-flow experiments using salt-water solutions, cylindrical obstacles, tracer particles, and laser-based flow visualization.
- Used MATLAB and PIVlab to extract velocity fields and analyze vorticity, velocity gradients, shear-layer structure, and wake dynamics across different gap widths and stratification strengths.
- Compared experimental regimes using Reynolds numbers, density contrast, and geometric confinement parameters.

Undergraduate Research Assistant — Stochastic Modeling and Genealogies

Apr. 2025 – Present

Advisor: Wai-Tong (Louis) Fan, UNC Department of Mathematics

Chapel Hill, NC

- Develop Python simulation workflows for stochastic population genealogies using `msprime`, `tskit`, NumPy, and pandas.
- Simulate and compare Wright–Fisher and multiple-merger ancestry models, including large-family and Dirac-coalescent-style genealogies.
- Analyze time to most recent common ancestor (TMRCA), merger structure, mutations, and site/branch frequency spectra across replicated simulations.
- Study how alternative stochastic reproduction mechanisms affect long-time genealogical structure and genetic variation.

Presentations

“Optical Inference of Density Stratification in Miscible Fluids via Refractive Signature Analysis”

Poster Presentation, UNC Office for Undergraduate Research

2026

Chapel Hill, NC

Honors and Fellowships

Summer Undergraduate Research Fellowship (SURF)

UNC Office for Undergraduate Research; \$4,000 competitive fellowship supporting independent research in applied mathematics.

2026

Kennedy Scholar — Mathematics Research Fellowship

UNC Department of Mathematics; \$2,000 fellowship supporting undergraduate mathematics research.

2026

Teaching Experience

Undergraduate Learning Assistant — MATH 110: Algebra

UNC Department of Mathematics

Aug. 2026 – Present

Chapel Hill, NC

- Support students during active-learning class activities through small-group and one-on-one mathematical problem solving.
- Provide Mathematics Help Center support in algebra, equations, functions, graph interpretation, and quantitative reasoning.

Selected Data Science Projects

Bobolink Migration and Climate Analysis

STOR 320

- Integrated more than 80 years of GBIF bird-observation records with ERA5 climate data to study long-term changes in migration-associated latitude and environmental conditions.
- Applied Python-based data cleaning, visualization, LOESS smoothing, regression, model comparison, and cross-validation while accounting for historical sampling bias.

FAIR-HIRE: Fairness in Automated Employment Decision Tools

Data Science Project

- Studied algorithmic fairness in automated employment screening, focusing on disparate impact, demographic parity, equalized odds, calibration, and lifecycle auditing.
- Synthesized technical, regulatory, and scholarly perspectives on how fairness metrics interact with legal and institutional constraints.

Leadership and Activities

UNC Badminton Club

Interim President (May–Aug. 2025); Treasurer (Aug. 2025–Present)

May 2025 – Present

Chapel Hill, NC

- Appointed interim president during a period of institutional and financial disruption; led an internal financial review and operational restructuring before transitioning to the treasurer role.
- Mixed Doubles D Runner-Up, 2026 Duke Open.

Technical Skills

Programming and Scientific Computing: Python (NumPy, pandas, msprime, tskit), MATLAB, R, Mathematica, Jupyter, L^AT_EX.

Numerical and Statistical Methods: ordinary differential equations, Euler and Runge–Kutta methods, numerical optimization, least-squares fitting, ray shooting, inverse-problem workflows, stochastic simulation, regression, cross-validation, and model comparison.

Experimental Methods: Particle Image Velocimetry (PIV/PIVlab), laser-based flow visualization, optical imaging, image processing, geometric and camera calibration, pixel-to-physical coordinate conversion, stepper-motor automation, fluid preparation, density stratification, and laboratory instrumentation.